

Nmap Scan Report

Ethical Hacking Task 02 – Network Scanning & Service Enumeration

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Tool Used: Nmap 7.94SVN

Objective

The objective of this scan was to examine my local machine using Nmap to identify open ports, running services, and basic network information. This exercise helped me understand the scanning and enumeration phase of ethical hacking.

Target Information

- **Target System:** Localhost
- **IP Address:** 127.0.0.1

Commands Used

nmap --version

nmap localhost

nmap -sV localhost

sudo nmap -O localhost

Scan Results

Host Status

- **Host:** Up
- **Latency:** 0.00010 seconds

Open Ports

Port Protocol State Service

22 TCP Open SSH

Closed Ports

- 999 TCP ports were closed (connection refused).

Running Services

Service Description

SSH Secure Shell (SSH) is a secure protocol used for remote login and remote system administration over an encrypted connection.

Operating System

Ubuntu Linux *(or replace this with the exact result shown by your nmap -O localhost command.)*

Observations

- The localhost system responded successfully to the scan.
- Only one TCP port (22) was open.
- SSH was the only detected network service.
- The remaining 999 TCP ports were closed, reducing the system's attack surface.
- No unnecessary network services were found during the basic scan.

Security Recommendations

1. Keep the SSH service updated with the latest security patches.
2. Use strong passwords or SSH key-based authentication.
3. Disable root login over SSH.
4. Configure a firewall to restrict SSH access to trusted users.
5. Close port 22 if remote access is not required.
6. Perform regular Nmap scans to monitor network services and identify unnecessary open ports.

Conclusion

The Nmap scan successfully identified the network services running on my local machine. The scan showed that only **port 22 (SSH)** was open, while the remaining **999 TCP ports were closed**, indicating a minimal attack surface. I learned that network scanning is an essential step in ethical hacking because it helps identify active services and potential entry points that could be exploited if left unsecured. I also gained practical experience using Nmap commands for host discovery, service enumeration, and operating system detection. This task improved my understanding of how security professionals assess systems while following ethical guidelines and scanning only authorized devices. Regular network scanning and proper configuration of services such as SSH help strengthen overall system security and reduce potential vulnerabilities.